

PROBLEM SET 2: TRANSITIONAL DYNAMICS IN MODELS WITH CONSTANT RETURNS TO ACCUMULATED FACTORS

PROBLEM I – THE AK MODEL WITH AN EXTERNALITY

We consider the following version of the Ramsey model:

$$\max \int_0^{+\infty} \ln C_t \cdot e^{-\rho t} dt, \quad (1)$$

$$Y_t = A_t K_t^\alpha L^{1-\alpha} \quad (2)$$

$$\dot{K}_t = Y_t - C_t \quad (3)$$

$$L = 1. \quad (4)$$

In this model, the level of technical progress A_t is *endogenous*. At each date t A_t is determined by an externality from the aggregate capital stock:

$$A_t = AK_t^{1-\alpha}.$$

Note that private agents do not take this relationship into account when making their decisions. Note also that the exponent of K in the externality is such that we have constant returns to capital in the aggregate. The aggregate production function is

$$Y_t = AK_t.$$

- 1 – Is the competitive equilibrium a Pareto-optimum in this setting?
- 2 – In order to obtain the equilibrium trajectory, can one compute the optimal trajectory of the economy by maximizing the utility of the representative agent under the physical constraints?
- 3 – Let A_t^* be the equilibrium trajectory for A_t . Show that the economy's trajectory is the competitive equilibrium of another economy (called B) where (1)-(4) holds but where instead technical progress is exogenous and equal to A_t^* .
- 4 – Is the competitive equilibrium a Pareto-optimum in economy B?
- 5 – Show how the preceding remarks may be used to solve for the competitive equilibrium of our economy.
- 6 – Show that the equilibrium conditions are such that

$$\frac{\dot{C}_t}{C_t} = \alpha A - \rho.$$

- 7 – Let $c_t = C_t/K_t$. Show that c_t follows a linear differential equation which is unstable in the usual sense.
- 8 – Show that if one rules out explosive solutions, c_t must be constant along the economy's trajectory.
- 9 – Show that the economy follows a balanced growth path from $t = 0$ on regardless of the initial value of K .
- 10 – What is the growth rate of the economy?
- 11 – Compute the trajectory of the social optimum.
- 12 – Is growth too high or too low in equilibrium compared to the social optimum? Why? Explain.

PROBLEM II – THE AUGMENTED SOLOW MODEL WITH ONLY ACCUMULABLE FACTORS

Consider the following model:

$$Y = AK^\alpha H^{1-\alpha},$$

where Y = output, K = capital, H = human capital

and:

$$\begin{aligned}\dot{K} &= s_K Y, \\ \dot{H} &= s_H Y.\end{aligned}$$

- 1 – Show that if (K_t, H_t) is an equilibrium trajectory for initial conditions (K_0, H_0) , then for any $\lambda > 0$ we have that $(\lambda K_t, \lambda H_t)$ is an equilibrium trajectory for initial conditions $(\lambda K_0, \lambda H_0)$.
- 2 – Show that the equilibrium ratio H/K converges to a unique long-term level. Compute this level.
- 3 – Compute the long-term growth rate of this economy.
- 4 – Under which conditions will the economy be on a balanced growth path from $t = 0$ on?